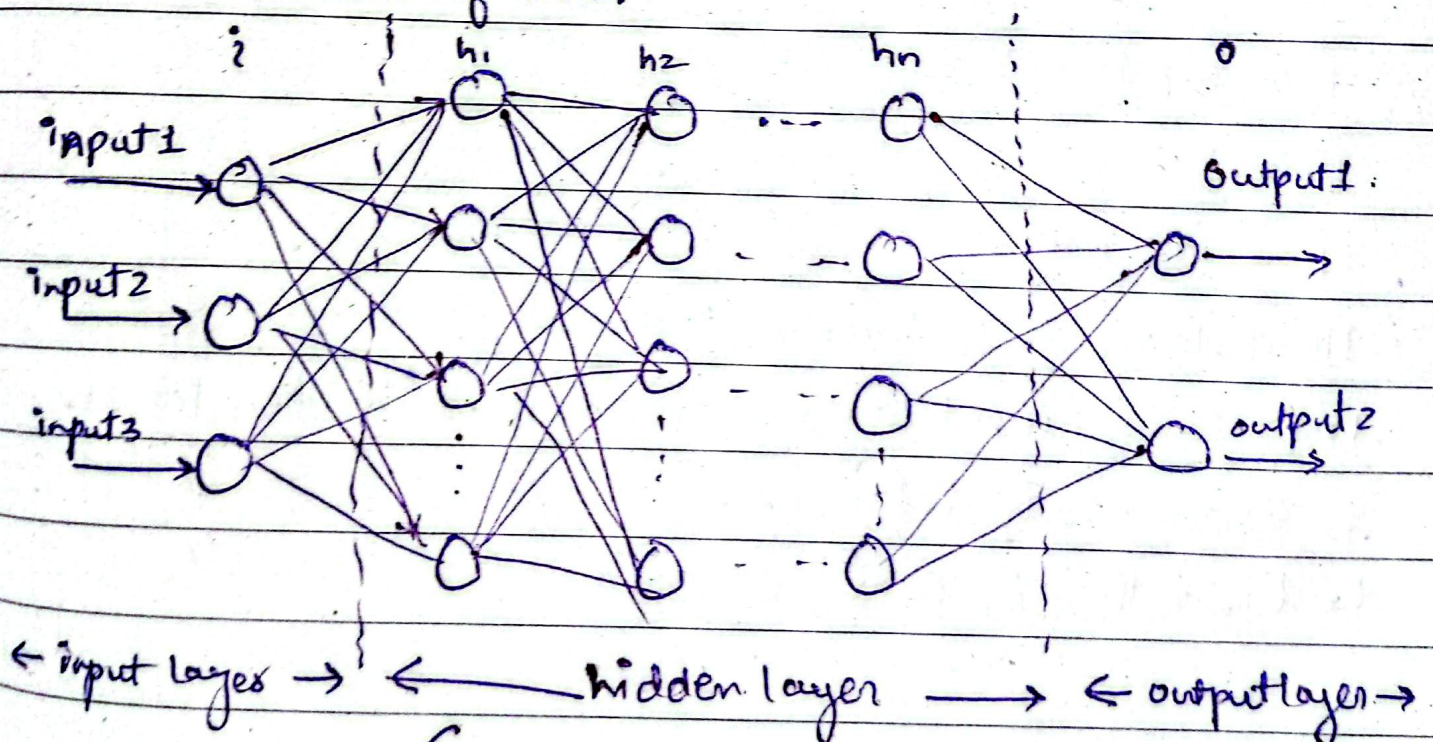


ARTIFICIAL NEURAL NETWORK (ANN)

- A data processing system consisting of a large no. of interconnected processing elements called artificial neurons (nodes). These neurons are connected thru connecting links.
- ANNs have self learning capabilities, allowing them to improve as more data becomes available.

→ Architecture of ANN



- ↳ not directly visible to external sys
- ↳ can be zero or more hidden layers
- ↳ responsible for intermediate computations & feature extraction

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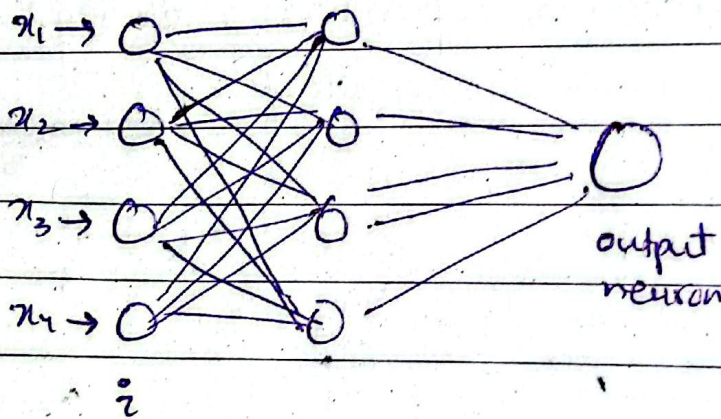
• Classification of ANN

① Single Layer Feed Forward Network

- Single Layer: Only one computational layer (output layer) making it a single layer ANN.

↳ Total two layers [input & output]

- Feed Forward: Information flows from input → output without any feedback loop.



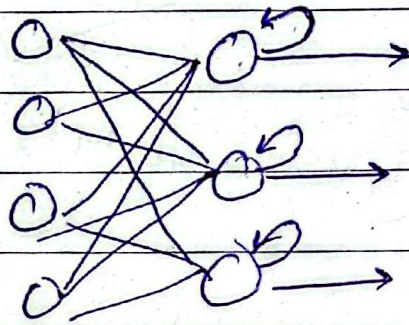
② Multilayer Feed Forward Network

↳ Contains input layer, one or more hidden layers and an output layer.

↳ Like the 1st example

③ Recurrent Network

- ↳ These network includes at least one feedback loop
- ↳ can be single / multilayer.



→ Vanishing Gradient Problem

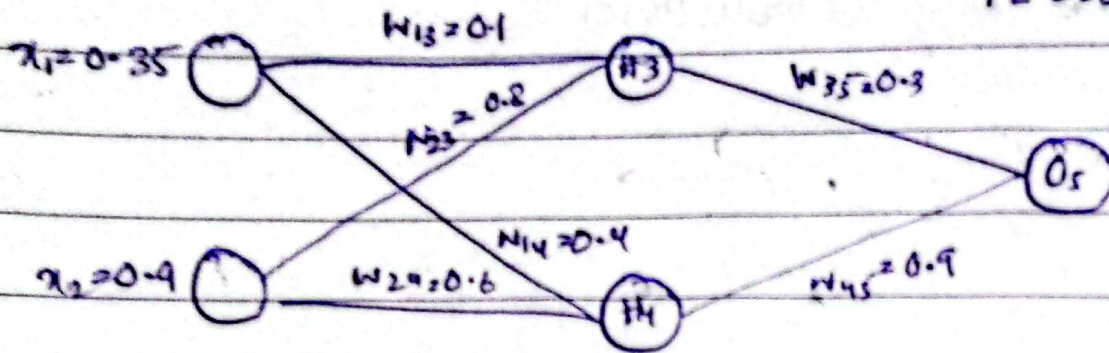
- ↳ This problem happens during backpropagation in deep neural networks when the gradient becomes extremely small as they move backward thru layers
- ↳ Because gradients become tiny:
 - ↳ weights update very slowly
 - ↳ early layers learn almost nothing
 - ↳ training becomes ineffective
- ↳ Mainly happens because of activation functions like sigmoid & tanh

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→ Backpropagation

learning rate = 1

 $Y = 0.5$ 

output

↳ For hidden layers we use Sigmoid Activation function
 & for hidden & " " " " $\tanh()$ " " " "

⇒ Forward Pass:

$$H_3 = x_1 \cdot W_{13} + x_2 \cdot W_{23}$$

$$H_3 = \tanh(0.35 \cdot 0.1 + 0.9 \cdot 0.8)$$

$$H_3 = 0.688$$

$$H_4 = \tanh(x_1 \cdot W_{14} + x_2 \cdot W_{24})$$

$$= \tanh(0.35 \cdot 0.4 + 0.9 \cdot 0.6)$$

$$H_4 = 0.591$$

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$$O_5 = S(H_3 \times W_{35} + H_4 \times W_{45})$$

$$= S(0.638 \times 0.3 + 0.591 \times 0.9)$$

$$O_5 = 0.673$$

For tanh

$$\hookrightarrow (1 - \tanh^2 h)$$

⇒ Backpropagation

For sigmoid

$$\hookrightarrow y(1-y)$$

~~use~~

↳ Output layer Error

For Sigmoid

$$\delta_5 = (\hat{y} - y) \hat{y} (1 - \hat{y})$$

$$= (0.673 - 0.5) (0.673) (1 - 0.673)$$

$$\delta_5 = 0.038$$

↳ Hidden layer Error

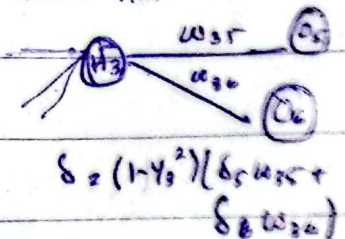
For tanh

$$\delta_3 = (1 - y_3^2) \delta_5 W_{35}$$

$$= (1 - 0.638^2) \times 0.038 \times 0.3$$

$$= 0.0067$$

$$\tanh'(x) = 1 - \tanh^2(x)$$

multiple hoti ho
sum hota

$$\delta_4 = (1 - y_4^2) (\delta_5 \times W_{45})$$

$$= (1 - 0.591^2) (0.038) (0.9) = 0.022$$

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$$b_{\text{new}} = b_{\text{old}} - n \cdot \delta$$

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↳ Weight updates

$$W_{ij} = W_{(\text{old})} - n \cdot \delta_j \cdot O_i$$

↓
piche
wale ka
error
↓
apne
oage
wale
ka
value

$$W_{35} = 0.3 - (1)(0.038)(0.638)$$

$$W_{35} = 0.275$$

$$W_{45} = 0.9 - (1)(0.038)(0.591)$$

$$W_{45} = 0.877$$

$$W_{13} = 0.1 - (1)(0.0067)(0.35)$$

$$= 0.097$$

$$W_{14} = 0.4 - (1)(0.022)(0.35)$$

$$= 0.3923$$

$$W_{23} = 0.8 - (1)(0.0067)(0.9)$$

$$= 0.793$$

$$W_{24} = 0.6 - (1)(0.022)(0.9)$$

$$W_{24} = 0.5862$$

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